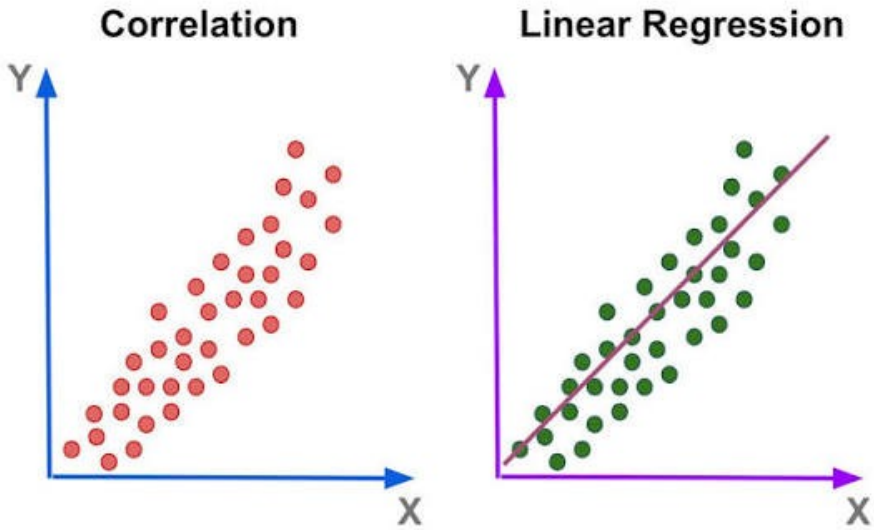
Lecture 25

Prediction I



Announcements

- Course project directed questions are out!

- Office hours every Tuesday (noon to 4pm) and Friday (11am to 1pm)
  - Link to book appointments is on the course website

Today

What does it mean to predict?

k-Nearest Neighbors

Types of Questions

Descriptive

- What does the data say?

Inference

- What can we infer about the population from the data?

Prediction

- Given some information, can we say what the next data point looks like?

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Inference

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- Are X and Y correlated in the population?

Prediction

- Given some information, can we say what the next data point looks like?
- If we know the value for X, can we accurately predict Y?

Delivery Robots

Suppose the dining halls ran a trial using delivery robots on campus

- Tracked delivery time, distance, items damaged, and snow depth
- Trial lasted for 300 deliveries



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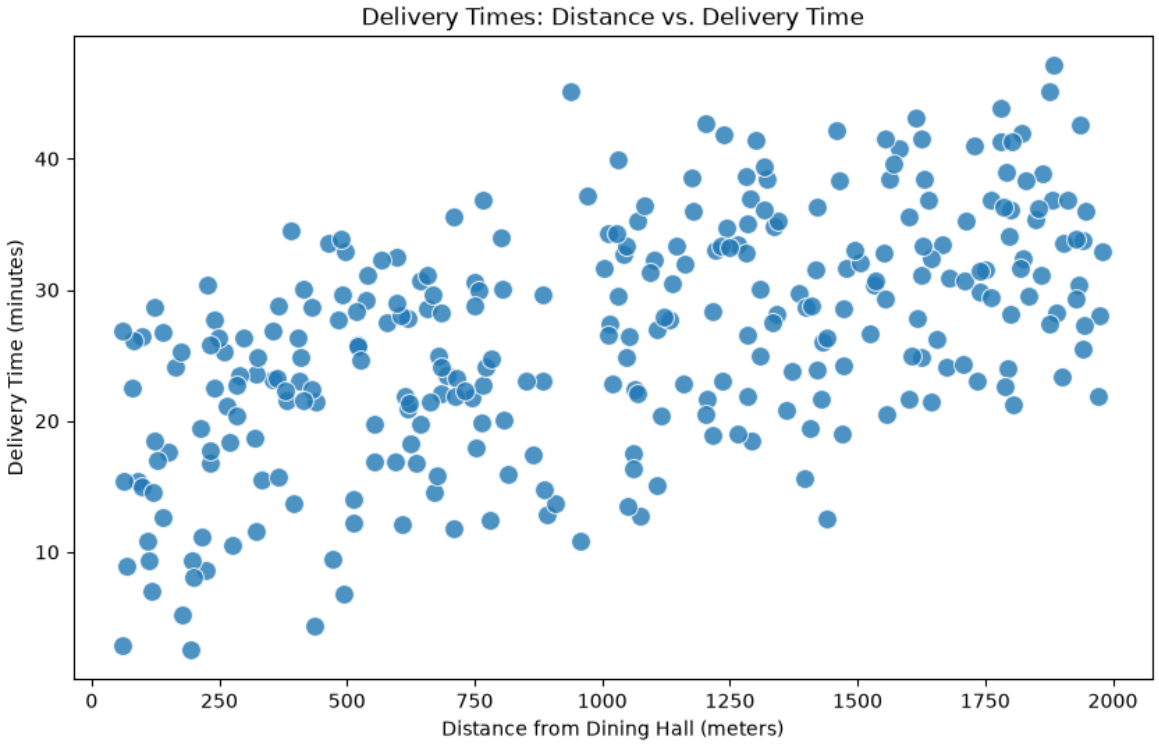
Using data from the trial, we're asked to

1) Predict how long a delivery will take
2) Predict whether or not items will be damaged

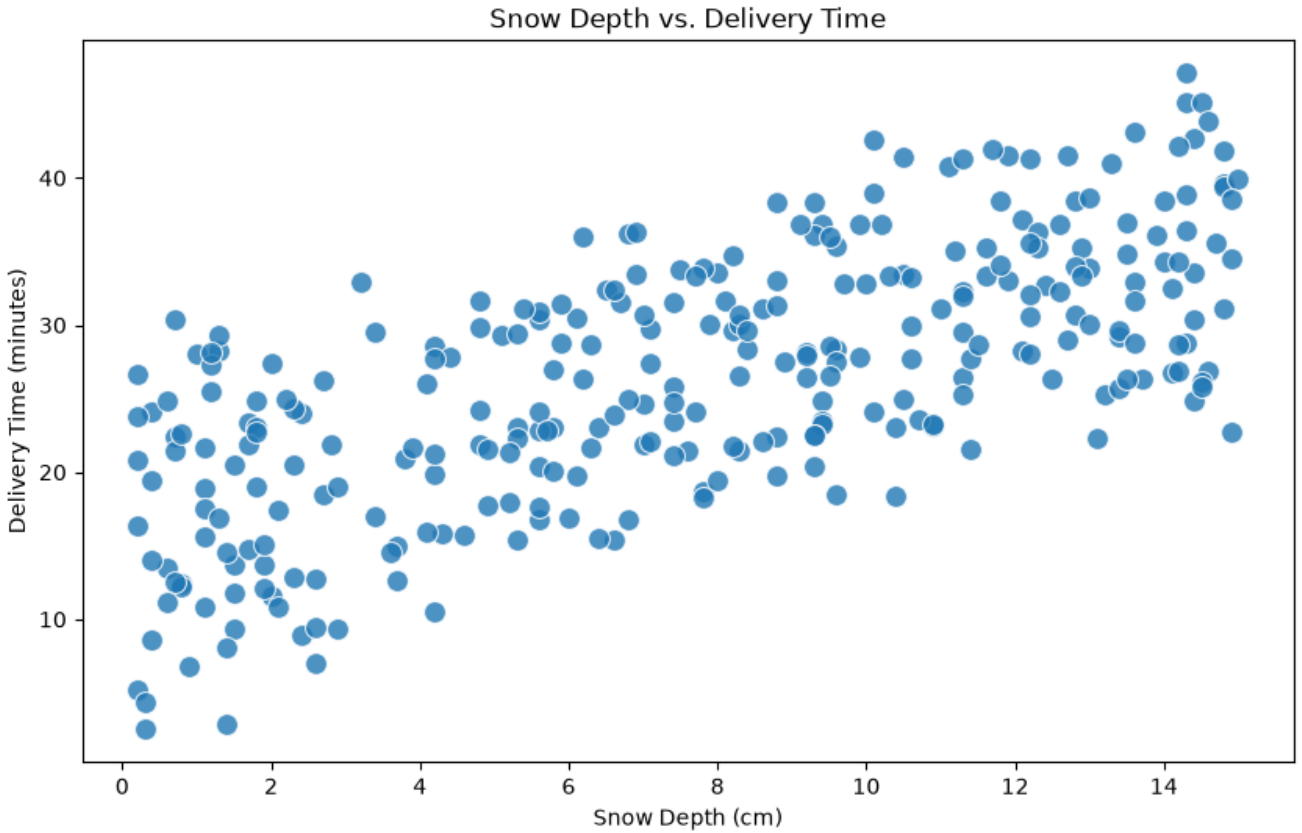
EDA


Distribution of Delivery Times
Count of Deliveries
0
5
10
15
20
25
30
10
20
30
40
Delivery Time (minutes)

EDA


Delivery Times: Distance vs. Delivery Time
Delivery Time (minutes)
10
20
30
40
0
250
500
750
1000
1250
1500
1750
2000
Distance from Dining Hall (meters)

EDA


Snow Depth vs. Delivery Time
Delivery Time (minutes)
10
20
30
40
0
2
4
6
8
10
12
14
Snow Depth (cm)

Language of Prediction

Variables

- X denotes inputs
- Y denotes output

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Goal: predict Y using X

- If Y is numerical, this is called **regression**
- If Y is categorical, this is called **classification**

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- X denotes inputs X = { distance, snow depth }
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Language of Prediction

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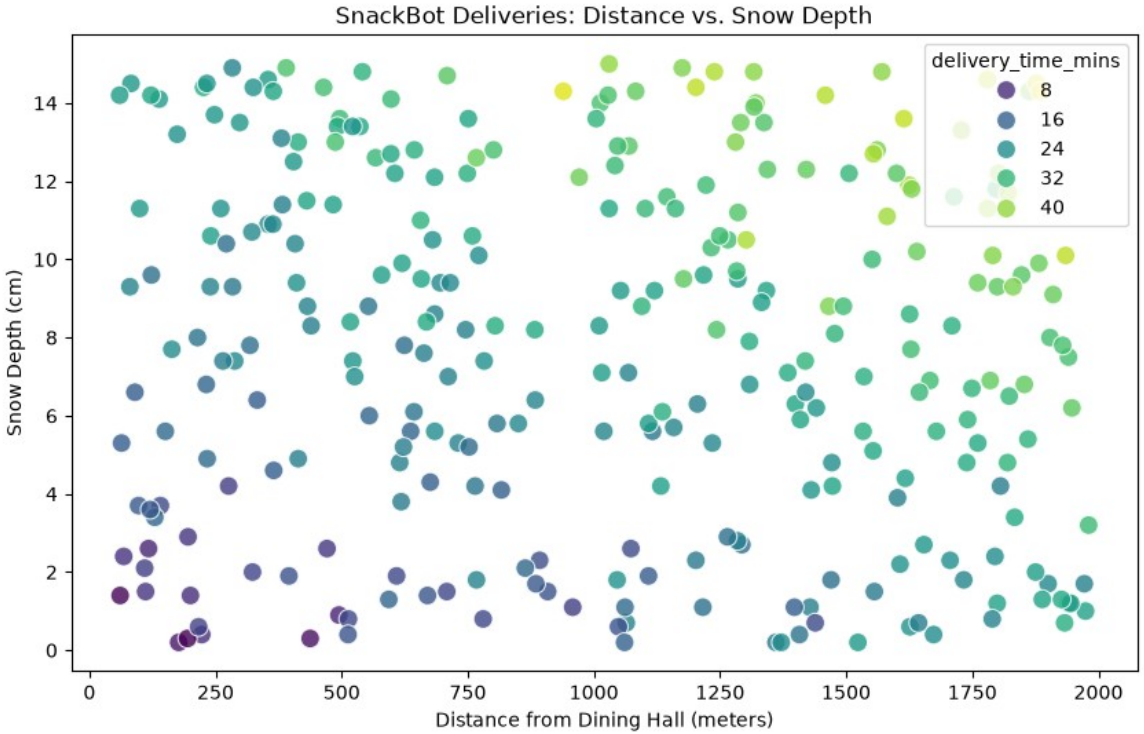
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Often call X the **feature space** and individual inputs as **features**

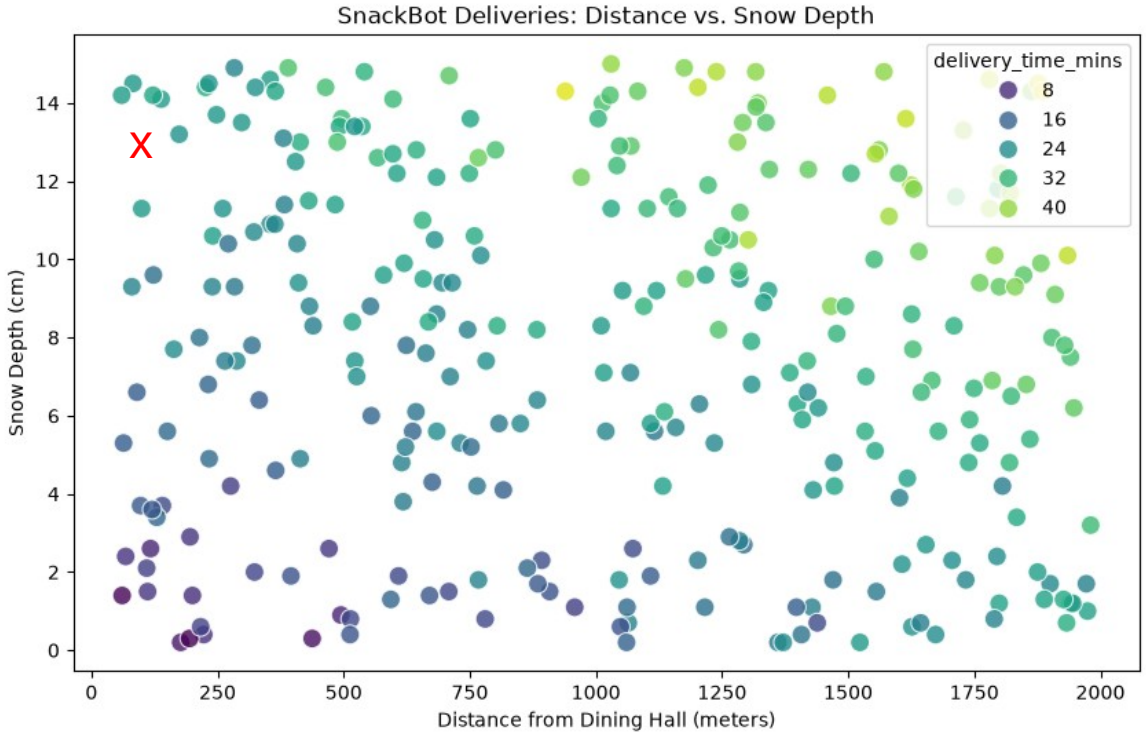
Nearest Neighbors (regression)



Nearest Neighbors (regression)

Let x be a new input value and let's predict its output

Idea: look at the Y values around x

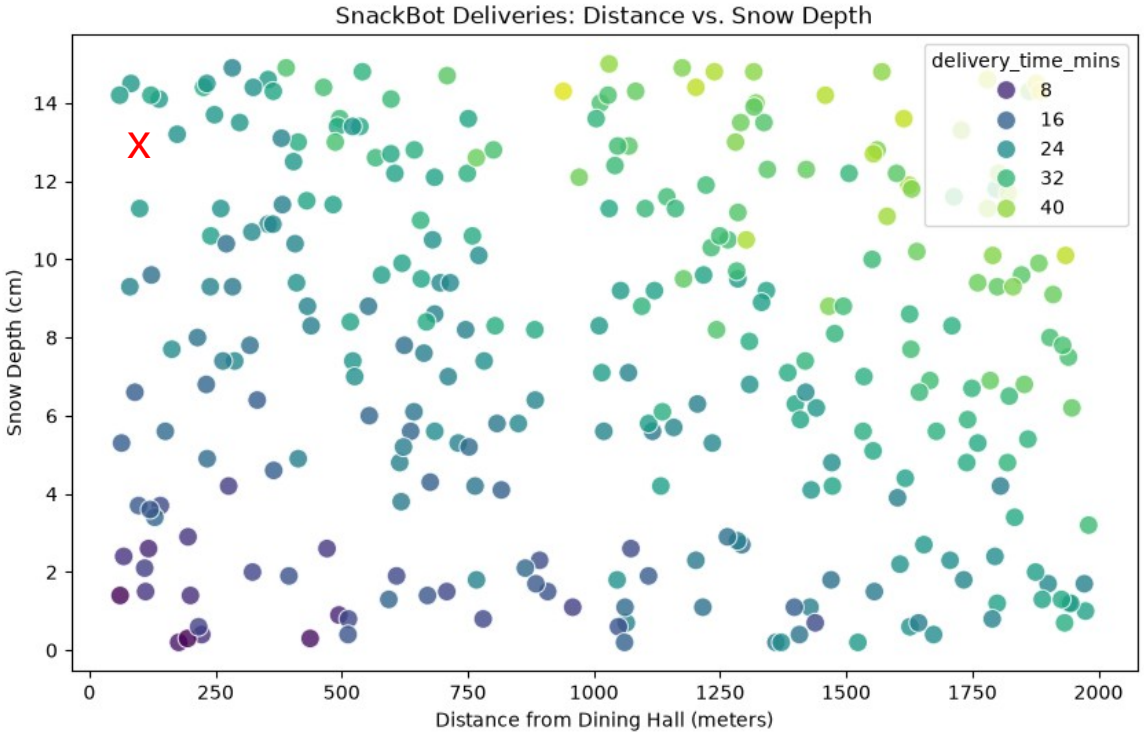


Nearest Neighbors (regression)

Let x be a new input value and let's predict its output

Idea: look at the Y values around x

1) Find the *k* nearest input values to x
2) Get each of their Y values
3) Average them together and output



Questions

What does nearest mean?

Questions

What does nearest mean?

- Is it straight line distance? Or something else?
- Should we scale the features? If so, how?

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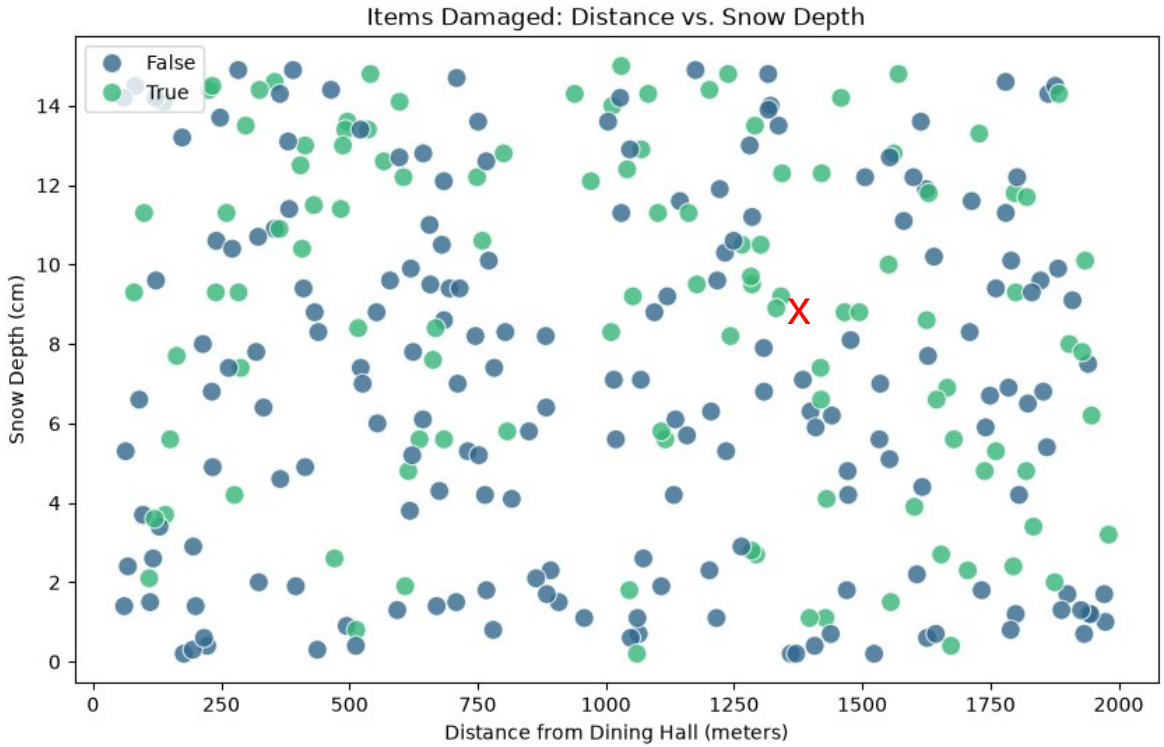
How do we choose k?

Nearest Neighbors (classification)

Let x be a new input value and let's predict its output

Idea: look at the Y values around x

1) Find the *k* nearest input values to x
2) Get each of their Y values
3) Output value with the most "votes"



Next time

- Linear regression